they show a large overlap with intrinsic motivation features, being independent of a specific external resource. Instead, they are functions of the agent-environment interaction history g and derived model M:

$$\zeta_j(s, a, s') = f_j(g, M) \tag{8}$$

for the jth appraisal variable. Note that the current state and action are actually included in g, but we emphasize that $f_j(\cdot)$ is not a function of the actual content of any state s (see Section 2.4 for a discussion of the extrinsic/intrinsic distinction). Rather, $f_j(\cdot)$ computes domain-independent characteristics, like 'recency' which may be derived from g, and 'motivational relevance' which can be derived by planning over M.

Intrinsic motivation is an active topic in developmental robotics (Oudeyer and Kaplan, 2007). Singh et al (2010) shows how incorporating these dimensions as extra reward provides better task achievement compared to non-intrinsically motivated agents (see Section 6.1). We discuss two implementations based on these ideas more extensively: Marinier and Laird (2008) and Sequeira et al (2011). The work by Marinier and Laird (2008) takes a diverse set of appraisal dimensions based on Scherer's appraisal theory (Scherer, 1999). These include both sensory processing dimensions, like suddenness, intrinsic pleasantness and relevance, and comprehension and reasoning dimensions, like outcome probability, discrepancy from expectation, conduciveness, control and power. The implementation by Sequeira et al (2011) uses a smaller subset of appraisal dimensions: novelty, relevance, valence and control. Note that these appraisal-based papers only elicit appraisal dimensions, without specifying categorical or dimensional emotions on top (see Table 9, i.e. appraisal papers with empty middle column).

We now highlight some appraisal implementations, both to concretize their specification in MDPs, and illustrate the differences between models. Sequeira et al (2011) specifies 'motivational relevance' as inversely related to the distance to the goal. If we implement a planning procedure over our model M which returns an estimated distance $\hat{d}(s, s^\circ)$ to the goal node s° from our current node s, then the associated appraisal variable for motivational relevance could be (Sequeira et al, 2011):

$$\zeta_{relevance}(s) = \frac{1}{1 + \hat{d}(s, s^\circ)} \tag{9}$$

Similarly, if we denote by $c(s)$ the number of time-steps since node s was last visited, then we can specify a 'recency' feature as (Bratman et al, 2012):

$$\zeta_{recency}(s) = 1 - \frac{1}{c(s)} \tag{10}$$

This example intrinsic motivation vector $\zeta = \{\zeta_{relevance}, \zeta_{recency}\}$ is used in Section 6.1 to show its use in reward modification.

There are several more specifications in intrinsic motivation RL literature that reflect appraisal dimensions. For example, Hester and Stone (2012a) maintain an ensemble of transition models (by stochastically adding new data to each model) and derive 'model uncertainty' from the KL-divergence (as a measure of the distance between two probability distributions) between the ensemble model's predictions:

$$\zeta_{uncertainty}(s,a) = \sum_{i \neq j} D_{KL}\Big[T_i(s'|s,a) \| T_j(s'|s,a)\Big] \tag{11}$$

for all pairs of models i and j in the ensemble. As a second example from their paper, 'novelty' of a state-action pair is identified from the closest L_1-distance to a historical observation:

$$\zeta_{novelty}(s,a) = \min_{<s_i,a_i> \in g} \|\langle s,a \rangle - \langle s_i, a_i \rangle\|_1 \tag{12}$$

Recently, Houthooft et al (2016) derive 'curiosity/surprise' from the KL-divergence between the old and new transition models (i.e. after updating based on the observed transition):

$$\zeta_{curiosity}(s,a,s') = D_{KL}\Big[T(\omega|g_t,a,s') \| T(\omega|g_t)\Big] \tag{13}$$

where $T(\omega)$ denotes the transition model parametrized by ω. Together, Equations 9-13 illustrate how intrinsic motivation and appraisal theory have modelled similar notions, and gives a short illustration of the variety of concepts that are expressible in the MDP setting.

It is also important to note that appraisal theory bears similarities to many 'domain-independent' heuristics developed in the planning community Russell et al (1995). These of course include heuristics without a clear psychological or biological interpretation, but we mainly emphasize the potential for cross-breeding between different research fields. For example, some appraisal theories partition novelty into three sub-elements: familiarity, suddenness and predictability (Gratch and Marsella, 2014). Each of these seem to capture different computational concepts, and such inspiration may benefit intrinsic motivation and/or planning researchers. Vice-versa, psychologist could seek for results from the RL or planning literature to develop and verify psychological theory as well.

There are several other implementations of appraisal dimensions, e.g. by Yu et al (2015), Lee-Johnson et al (2010), Williams et al (2015), Si et al (2010), Kim and Kwon (2010), Hasson et al (2011) and Moussa and Magnenat-Thalmann (2013). We also encounter a few explicit social dimensions, like social fairness (Yu et al, 2015) and social accountability (Si et al, 2010), although the latter for example requires some symbolic reasoning on top of the RL paradigm. This illustrates how current RL algorithms (for now) have trouble learning complex social phenomena. Some of the appraisal systems also include homeostatic variables (Yu et al, 2015). Both Williams et al (2015) and Lee-Johnson et al (2010) do not mention appraisal in their paper, but their dimensions can be conceptualized as intrinsic motivation nevertheless.

In summary, some appraisal-based dimensions require cognitive reasoning, and are harder to implement. However, dimensions like novelty, motivational relevance and intrinsic pleasantness are frequently implemented (see Table 3). Table 4 provides a more systematic overview of the actual connections to the RL framework. These features usually require learned transition functions, recency features or forward planning procedures over the model space, which can all be derived from the history g. Also note that a single concept may be interpreted in very different ways (Table 4). For example, control and power have been derived from the transitions function (Kim and Kwon, 2010), from the number of visits to a state (Sequeira

Table 3 Overview of frequently investigated appraisal dimensions.

Appraisal dimension	**Paper**
Novelty	(Sequeira et al, 2011) (Kim and Kwon, 2010) (Si et al, 2010) (Williams et al, 2015)
Recency	(Marinier and Laird, 2008)
Control/Power	(Marinier and Laird, 2008) (Sequeira et al, 2011) (Kim and Kwon, 2010) (Si et al, 2010) (Williams et al, 2015)
Motivational relevance	(Marinier and Laird, 2008) (Sequeira et al, 2011) (Hasson et al, 2011) (Kim and Kwon, 2010) (Si et al, 2010) (Williams et al, 2015)
Intrinsic pleasantness	(Marinier and Laird, 2008) (Sequeira et al, 2011) (Lee-Johnson et al, 2010)
Model uncertainty	(Marinier and Laird, 2008) (Lee-Johnson et al, 2010) (Kim and Kwon, 2010) (Williams et al, 2015)
Social fairness/attachment	(Yu et al, 2015) (Moussa and Magnenat-Thalmann, 2013)
Social accountability	(Si et al, 2010) (Kim and Kwon, 2010)

Table 4 Overview of the five most frequently investigated appraisal dimensions (columns) and their specific implementations in six appraisal-based papers (rows). The cell text indicates which event causes the associated appraisal dimension to be high. Note that both Williams et al (2015) and Lee-Johnson et al (2010) do not explicitly mention appraisal theory as their inspiration, but they do derive emotions from dimensions encountered in appraisal theory. Only the implementation of Marinier and Laird (2008) uses direct sensory information (for control and intrinsic pleasantness), which would better fit with the hard-wired approach in Section 4.4. All other specifications rely on (an aggregate of) the agent-environment interaction history, for example on an estimated transition model $T(s'|s,a)$.

	Novelty/ Suddenness	**Control/ power**	**Motivational relevance**	**Intrinsic pleasantness**	**Model uncertainty**
(Kim and Kwon, 2010)	Ratio of $\sum_{s'} T(s'\|s,a)^2$ and $T(s'\|s,a)$	Entropy reduction by act sel.	High TD	-	Low belief $b(s)$ & high goal distance
(Lee-Johnson et al, 2010)	-	-	-	Low mean travel time	Mismatch of model and obs.
(Marinier and Laird, 2008)	High time to last state visit	Absence of obstacles	Low dist. to goal	Absence of obstacles	Low progress
(Sequeira et al, 2011)	Low # visits to state	High # visits to state	Low dist. to goal	Current reward/ value ratio	-
(Si et al, 2010)	Low T of obs. transition	Low dist. to higher value state	High absolute TD	-	-
(Williams et al, 2015)	Unseen/seen ratio state-space	High success/fail ratio	Part of task finished	-	Low model accuracy

et al, 2011), from a forward planning procedure (Si et al, 2010) and from the overall success of the agent (Williams et al, 2015). We encounter a fundamental challenge in the field here, namely how to translate abstract cognitive concepts to explicit (broadly accepted) mathematical expressions.